

COURTSIDE: trading the seconds after a tennis point

Gator Quant Hacks 2026 · Systematic Trading · Repo: github.com/OjasMishra32/courtside · Reproduce: `python run_all.py --oos, python scripts/v2_burned_oos.py`

1. Economic foundation

A tennis point ends in tiers. Ball tracking knows where the ball will land while it is still in the air; the umpire knows at the bounce; licensed data feeds, and the market makers on them, know about a second later; TV, streams and score widgets know tens of seconds later. Match-win probability is a known function of the score, and we built the exact point-level Markov chain (`src/markov.py`). So every point moves fair value by a computable amount, its **leverage**. A simulated ATP best-of-3 has ~161 points with mean $|\text{leverage}|$ 5.7%, and fair value travels \$4.20 per share over the match.

Who is on the other side? Whoever is acting on an older tier. A trader who knows the point before the book reprices buys from a stale quote; a trader who acts after it pays the spread to someone faster. The gaps are physical (camera frame rates, data licensing, stream delays), so they persist. Polymarket's own design admits it: sports markets hold every marketable order for 1 s (3 s before May 2026) so makers can reprice. H1–H4 were committed before any result (`HYPOTHESIS.md`, commit 7232986). H5–H6 were written after in-sample results and frozen before OOS. v2 and its blind forward test are in `HYPOTHESIS_V2.md`, and every change is in `DEVIATIONS.md`.

2. Data and method

- **Every resolved ATP/WTB singles moneyline on Polymarket with $\geq \$5k$ volume, Oct 2025–Oct 2026: 13,084 matches, \$2.84B traded.** Full taker tapes come from the public data API. Each print is classified as lifting the ask or hitting the bid (buying one outcome is selling the other), so fills pay the spreads that were actually traded. Every match is charged its own fee ($\text{rate} \cdot q(1-q)$ per share; rate 0, 0.03, then 0.05) and its own order delay. We measured that tape timestamps are on-chain block times, a median **1.98 s after the true match time** (5,472 trades joined to the live websocket by transaction hash).
- **In sample:** the first 80% of matches. **Out of sample (OOS):** the last 20%, from 2026-08-25, locked until v1 was frozen and opened once for v1. v2 was built knowing v1 lost there, so that window is burned: v2 is shown on it labelled non-blind, and every look is logged (`results/oos_peeks.log`). **Forward:** matches that start after 2026-10-03 13:00 UTC, the blind test for v2.
- **Live, on 2026-10-03:** we recorded Polymarket order books (every tennis and table-tennis market) alongside four score sources, with millisecond receive times.
- **Tracking:** a physics Monte Carlo of Hawk-Eye-class tracking (340 fps, ± 3.6 mm); real 120 fps table-tennis video (OpenTTGames); broadcast tennis (TrackNet). Processing ran on HiPerGator.

3. What the data says

Table 1. Tests, net of fees and traded spreads (¢/share; 95% CI clustered by match).

Test	In sample	Out of sample (v1 run, opened once)	Verdict
H1 Follow the jump after the delay	−1.61 [−1.65, −1.57]; all 20 variants −1.5 to −1.9	−2.08 [−2.17, −1.98]	Fails
H2 Buy favourites entering 0.85–0.97	−0.41 [−1.44, 0.59]; live prices calibrated within ~1¢	+0.80 [−1.36, 2.66]	No edge
H5 Maker quoting after jumps (written after IS)	+0.22 [−0.01, 0.43]	+0.16 [−0.49, 0.87]	Inconclusive
H6 Fast tier, 30 s markout, walk-forward (written after IS)	+0.6 to +2.4, 8/8 months > 0	+0.4 to +0.8, 3/3 months > 0	Holds

Test	In sample	Out of sample (v1 run, opened once)	Verdict
Everyone else in the same 0–3 s window	–0.5 to –1.7, every month	–1.2 to –1.9	
Copying the fast tier 3 s later	< 0 every month	< 0 every month	Edge is speed

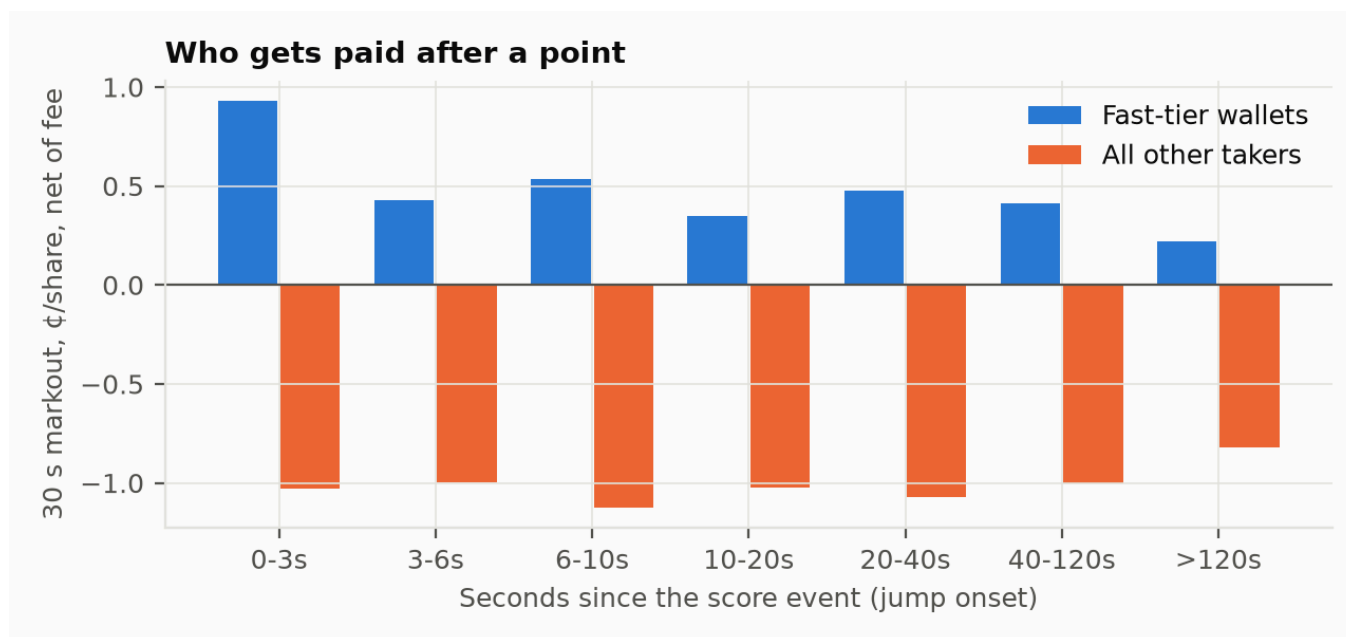


Fig. 1. 30 s markout per taker print, net of fee, by seconds since the score event. Fast-tier wallets are qualified each month using earlier months only. Onset-aligned ex-post event study; tradable numbers (Table 2) use the causal window.

Chasing the move loses, and live prices are calibrated, so slow money has no edge. What exists is a **persistent fast tier**: wallets that trade within 3 s of a score event and beat the market every month, in sample and out. The other side of their trades is the ~1¢ a share that everyone else loses. Copying them with a lag loses too, so the edge is speed and cannot be followed.

4. The strategy (v2) and its evidence

v1 (the fast tier's trades, ≤\$1k each, held to resolution) earned +1.16¢/share in sample but **lost \$36k out of sample**. Big tickets on cheap tokens made its P&L a lottery. Six optimisation lenses then ran on in-sample data only, each checked by two adversarial verifiers (`research/v2/`). The survivors are frozen as v2 (`HYPOTHESIS_V2.md` , `src/v2.py`):

1. Trade the walk-forward fast-tier opportunity set (prints inside 0–3 s of a score event).
2. Keep a wallet only if its shrunk past edge beats **today's** fee: $0.05 \cdot q(1-q)$.
3. **Risk-parity size**: shares $\propto 1/\sqrt{q(1-q)}$, capped by the print and at \$1k.
4. **Cap net exposure at 100 shares per match**; trades that reduce risk are always allowed.
5. **Hold to resolution**. No exit orders.

Refuted and dropped: **maker exits**. They looked like Sharpe 10 on the tape, but the verifiers found a wrong tick size. On live books a resting exit after a jump filled only 35% of the time within a minute. When it did fill, the price kept running another 5.6¢; when it did not, the price fell 5.7¢. That is adverse selection.

Table 2. v2 (causal window), net of fees, held to resolution. The 0–3 s window is measured from the moment our detector could fire, not from the onset seen in hindsight. A verifier found the onset labelling put 81% of the first v2 draft's OOS P&L on trades made before detection (D9).

	In sample, Feb–Aug 2026	Burned OOS, Aug 25–Oct 3 (non-blind)	Forward, from Oct 3 14:00 UTC (blind)
Trades	55,662	10,412	[FWD_N]
Net per share, at fast-tier fills	+1.38¢ [1.17, 1.59]	+0.60¢ [0.09, 1.13]	[FWD_RES]
... with +½ tick (0.5¢) worse entry	+0.88¢ [0.67, 1.09]	+0.10¢ [−0.41, 0.63]	
... with +1 tick (1¢) worse entry	+0.38¢ [0.17, 0.59]	−0.40¢ [−0.91, 0.13]	
P&L / capital (4 h lock per position)	+\$40.4k / \$28.3k	+\$3.7k / \$22.8k	[FWD_PNL]
Sharpe / max DD	14.5 / −2.0%	6.7 / −2.1%	
Months positive	7/7 (7/7 even at +1 tick)	2/3 (Oct = 3 days)	
Forward primary A: fast tier – others, 30 s net			[FWD_A]
Forward primary B: v2 book, 30 s net per share			[FWD_B]

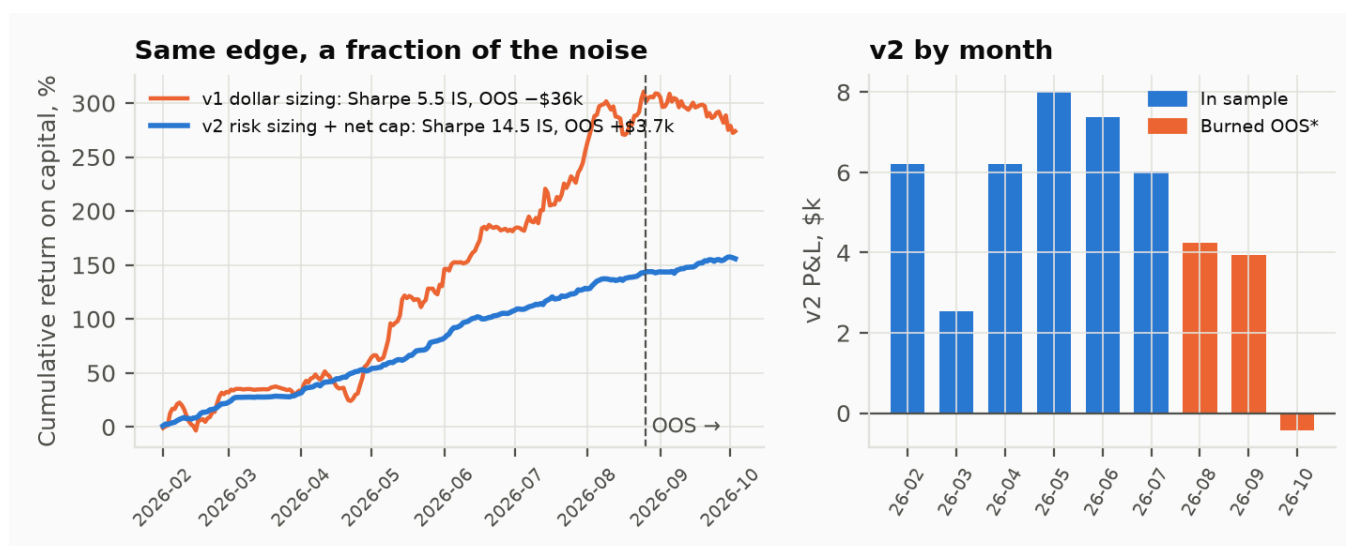


Fig. 2. v1 vs v2, return on each book's own capital, and v2 P&L by month (*Aug includes IS days).

Table 2 is a **paper book on the fast tier's own fills**: it prices the opportunity at their speed, not our execution, which needs in-venue tracking plus a co-located gateway. **Reading Table 2.** In sample the edge survives a full tick of slippage in every month. In today's regime (1 s delay, 5% fee, 131 qualifying wallets) it is positive only at the fast tier's own fill prices and is gone at half a tick. The strategy pays whoever is *first* to the stale quote, and the margin for second place is now under half a tick. That is the case for in-venue ball tracking plus a co-located gateway (Section 5), and the reason a remote copy cannot work. The Sharpe is high because v2 places ~270 small, nearly independent bets a day, each held to an exogenous binary outcome, with net exposure capped per match. All nine net-cap and deployment settings in the sizing study have per-share CIs above zero (onset labelling), so this is a plateau, not a tuned spike.

5. Latency: who can actually be in the fast tier?

Signal	When it knows the point, vs the official WTA point timestamp
Ball tracking, Hawk-Eye class (simulated: 340 fps, ±3.6 mm)	100–300 ms <i>before the bounce</i> : 1 SD landing error ±2.4 cm at 100 ms, ±10.7 cm at 300 ms
Ball tracking, real 120 fps video (table tennis)	misses called 50 ms before contact : 11 of 11 calls correct (11 of 41 misses called, recall 27%); median first-call lead 25 ms

Signal	When it knows the point, vs the official WTA point timestamp
Ball tracking, 25 fps broadcast (tennis)	useless: 0.7–1.2 m median landing error
Polymarket book (market makers)	–1.2 s (reprices before the official stamp) , $n = 482$ points
Kalshi	reprices with the market makers; leads Polymarket on 69% of 106k repricings by ~2 s, mostly Polymarket's own order delay; $6.1\times$ its in-play volume
ESPN / Polymarket sports feed / WTA API	+27.5 s / +29.1 s / +43.3 s; 0 of 295 changes beat the book by > 1.3 s

Market makers already sit on the official feed, and Kalshi is where prices are discovered. **Remote traders on public data are the slow tier.** No public feed beats the book, and a Kalshi→Polymarket laggard trade nets ≈ 0 today. The fast tier needs a signal that arrives before the official feed: in-venue, high-frame-rate ball tracking. Our results show where that lead comes from (frame rate and camera count) and how large it is (tens to hundreds of ms on top of a ~1 s official-feed lag). Infrastructure: Polymarket's API sits behind Cloudflare's Miami edge, with a 67 ms median one-way feed delay from Gainesville and an origin consistent with London. Co-locating there saves ~130 ms per round trip, and that decides queue order behind the 1 s delay.

6. Risk management

- **Venue rules (largest risk).** Moving to a 5% fee cut the fast tier's edge by about two-thirds. A longer delay would cut it again. v2 re-prices its wallet filter at the live fee every month. We halve size if the trailing-month net edge falls below 0.3¢ and stop at ≤ 0 .
- **Crowding.** Qualifying wallets grew $4 \rightarrow 131$. Edge per share fell from ~2.4¢ to ~0.8¢ but stayed positive every month.
- **Position limits.** Net $|\text{exposure}| \leq 100$ shares per match; $\leq \$1\text{k}$ per order; capital = $3\times$ peak locked. The worst historical match lost \$206 (v1: \$3,098); the worst day was -1.9% in sample and -2.1% out.
- **Wrong calls and outages.** Trade only calls with $P \geq 0.95$. Kill switch on any feed or tracking dropout over 2 s, and on measured order latency beyond its 95th percentile.
- **Resolution.** 2.9% of matches settled 50/50 (retirements); this is included in all P&L.
- **Legal and access.** Courtsiding breaks most tournaments' ticket terms, live tracking data is licensed, and Polymarket's international venue restricts US persons. A deployment needs licensed data, a permitted venue and legal review. This repo only reads public data and never places orders.

7. Liquidity and capital deployment

- **Tennis is deep:** median 1¢ spread, \$8.1k at the touch and \$61k within 2¢ (live sample). **Table tennis is not tradable on Polymarket:** 89¢ median spread, \$23 at the touch, ~\$2 of volume per match.
- **What there is to take, per point (live, 482 points).** Depth resting at prices the next reprice makes stale: median \$222–565 (mean \$1.3–3.1k) from 2 s to 0.25 s before the reprice, and gone 0.5 s after it. Even an oracle that knew every point and held to resolution would net a mean +\$18–21 per point at today's fee (median $\approx \$0$; +\$38 on moves $\geq 3\text{¢}$). Exiting at the touch loses after the second fee. With the 1 s order delay, an order must leave ≥ 1.3 s before the reprice, ≈ 2.5 s before the official point stamp: that is the time budget a tier-0 signal has to beat. Prints landing in the 0.5 s before a reprice were 98% with the move and earned +0.91¢/share net, so some takers already have the point 1–1.5 s before the book.
- **v2 is small by design.** ~\$1.48M traded over 206 days on \$28k capital (turnover $\sim 93\times$ a year; peak locked \$9.4k). Capacity is bounded by the stale quotes resting at each point (a few \$k at the touch) and shared with the existing fast tier. Fast-tier volume in the 0–3 s window ran \$0.3–3.1M a month. Side markets add only ~\$2k/month.

8. Variants and caveats

Strategy variants backtested: 44 (H1–H6) plus 3,342 across the six v2 lenses (exit 902, sizing 111, selection 341, cross-market 68, latency 16, Kalshi 1,904). All are reported in [results/](#) and [research/v2/](#), along with their verifier reports. The expected best Sharpe from luck over this many trials is ~ 4.8 (Bailey & López de Prado); v2 shows 14.5 in sample and 6.7 on the burned OOS. The v2 forward test is the clean check. **The v2 book trades the fast tier's own fills.** It measures the opportunity at that speed, not our execution, which would need in-venue tracking plus a co-located gateway. H1–H6 use onset-aligned windows as an ex-post event study. Every tradable number (v2) uses the causal window. References: Bailey & López de Prado (2014); Harvey, Liu & Zhu (2016); Klaassen & Magnus (2001); Voeikov et al. (2020) TTNNet/OpenTTGames; Huang et al. (2019) TrackNet; Polymarket and Kalshi API docs.